

Optical Based Non Invasive Glucometer with IoT

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Abstract— Diabetes mellitus (DM) nicknamed as sugar is becoming one of incurable and critical challenge to medical field. Serious illness such as premature mortality even cause of death, which can be identified by continuous monitoring from a remote location. The traditional and conventional method such as finger pricking's demerits such as pain and damage to tissues that cause infection. Such type of infections can be overcome by non invasive technique. The objective of this paper is to monitor glucose level using non invasive technique by optical and IoT technology. The proposed sensor circuit consists of IR LED's of wavelength 650-2500nm for optical blood glucose measurement and NIR photodiodes (InGaAs) to receive the reflected light from body parts to determine the glucose level. The Beer-Lambert law is used for signal processing along with GSM based IoT real time information transmission. The project is implemented using Arduino IDE for finding the performance matrix of the system as well as various analytical studies.

Keywords— Glucose level, Non invasive, optical method, NIR LED, NIR Photodetector, IoT.

I. INTRODUCTION

Diabetes mellitus is a medical condition in which body remains in improper insulin production and regulation. Insulin hormone act as a key allow glucose to enter into body tissues to maintain blood glucose level in blood. Diabetes is also known as sugar are of mainly 2 types, Type1 known as insulin dependent and have 5-15% of all cases. Type 2 non-insulin dependent occurs in childhood and insulin dose to have a healthy life. Type2 can be controlled by regular exercise, medication and a nutritious diet. The type 2 category mainly seen in people of 40+ age due to irregular food and health habits.

Blood glucose measurement allows detection of diabetic conditions in ICU, ER and operation theatres. Diabetes leads to serious illness such as brain dysfunction, eye irritation, kidney-heart failures and even premature mortality. Unfortunately, there is no any permanent cure for diabetes. However, continuous monitoring is one of the solutions by seeking medical help. in finger pricking pain, infectious transmission of diseases, bio-hazardous waste and damage to finger tissues may cause. To avoid demerits of conventional method idea of non invasive measurement arises. Also non invasive method reduces the need of trained personnel for

glucose monitoring. As per survey about 3 times a day diabetes measurement is needed to control and regulate it. Since, by world health organization by the year of 2030, diabetics reach to 366 million from 171 million current scenarios

NIR light found to penetrate biological tissues at a depth of 1-10mm developed a method for biomedical and clinical sensing. The absorption of light in the whole specimen provides an idea of glucose level. Several methods are used to measure and predict glucose level. A number of optical technologies such as electromagnetic sensing, near infrared spectroscopy, Raman spectroscopy, reverse iontophoresis, Fourier transforms, and infrared spectroscopy has been invested and utilized. Most common methods analyze absorbance of light and utilize spectrophotometric study based on the Beer-Lambert law.

The Beer Lambert law is combination of two laws, (a) the intensity of transmitting light decreases exponentially as concentration of substance increases, (b) the intensity of transmitting light decreases exponentially as distance travelled through substance increases.

Most of biological cells and fluids are transparent in wavelength 700-1100nm. The optical window of tissues helps to identify glucose level. The glucose molecule C₆H₁₂O₆ consists of C-O, O-H and C=O bonds, The presence of these bonds causes absorption of NIR light than other human bodily fluids in blood. This technique used for blood monitoring as water content avoided regions for reducing interference.

On healthcare applications recent implementation utilizes it based as well as smart mobile devices for communication gateways. it provides remote and real time push notifications to the patients and their concerns persons.

A. EXISTING NON INVASIVE METHODS

A prediction of treatment algorithms gives the accuracy to the measured value. Different optical methodologies are available. Investigations mainly described are polarimetry, metabolic heat conformation, ultrasound, thermal emission, photo acoustics, Raman light absorption and transmittance spectrums. [1]

1. Light absorption spectroscopy

When light hits on biological tissues of the human body due to reflection, transmission and scattering a proportional variation occurs with the structure and chemical components of the sample.

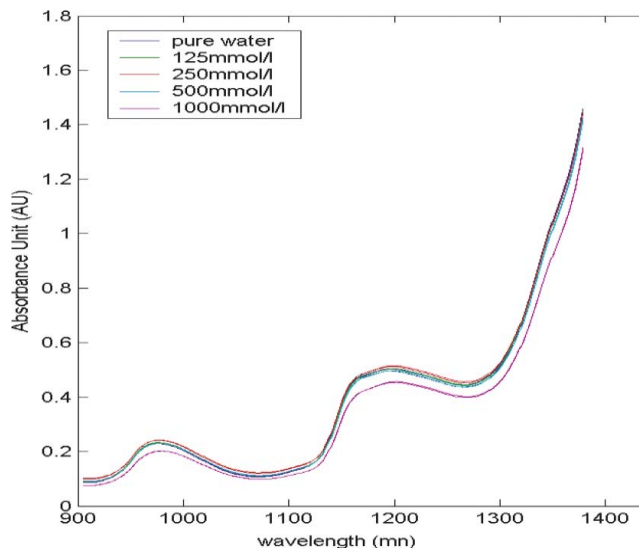


Fig.1: NIR Spectra-Glucose solution absorbance against wavelength. [1]

Main regions of spectroscopic investigations are done in visible and near infrared (NIR) range, namely around 590nm-1180nm. Middle range (MIR) propagates into few micrometers and applied to blood samples observed, whereas NIR get penetrated more than MIR.

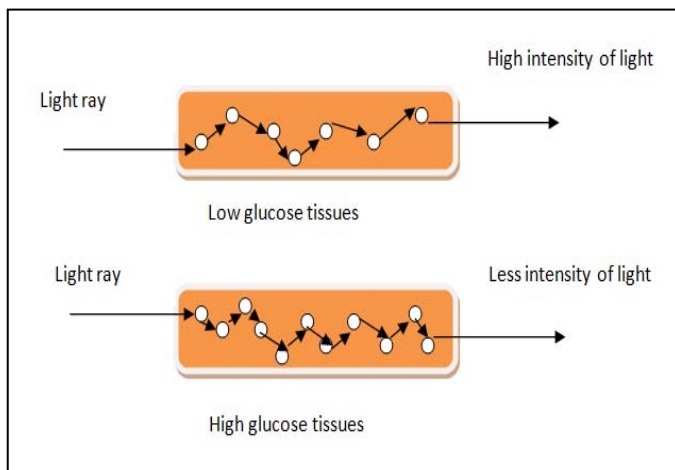


Fig 2: Scattering of light through tissue [2]

Glucose produces weakest NIR absorption signal per concentration unit of body composition. NIR measures a depth of 1 to 100 millimeters. When the wavelength value increases penetration value decreases. Ear lobe, finger cuticle and skin are the main body parts used for observations.

2. FIR Spectroscopy

Body heat and thermal emission are used for glucose detection. It is a radiation technology, which needs no any external energy source. [5]

3. Raman spectroscopy

Due to oscillations and rotation of transmitting light get scattered and its value varies. It is tested with blood, water, serum and plasma solutions. The instability in intensity of laser wavelength causes error due to tissue chemicals. [5]

4. Photoacoustic spectroscopy

Rapid heat formation due to optical beams and generates an acoustic pressure wave, measured using microphones. An ultrasonic wave is generated as a part of the absorption of pulsating light. Pressure variations arise due to heat variations when laser beams hits on biological tissues. [5]

5. Polarimetry

Linear polarization of light with respect to path characteristics, temperature and concentration is considered. A clear optical depolarization of the beam occurs in aqueous humor of the eye.

6. Fluorescence

Fluorescence is a sensing technology carried with tears by painless method. The photonic sensing is done by polymerizing crystalline arrays with respond to different concentration of diffraction of visible light.

II. METHODOLOGY

To obtain glucose concentration, using glucose spectroscopy between wavelengths 940 nm to 2450 nm. To setup system for transmission and reception of NIR rays, a reflective optical sensor is used with the fingertip as the body site. As light is transmitted by infra red sensor which will fall on earlobe/fingertip as we have taken finger as a body site. The reflected light is converted into voltage by photo detector and by reflectance spectroscopy. This voltage signal is then processed for signal conditioning before feeding into the microcontroller. These signal conditioning parts consist of filters and amplifiers. This filtered and amplified signal is fed into microcontroller at analog pin A0 for converting it to digital form. Finger tip is placed between NIR light source and NIR photo detector as shown in figure 3.

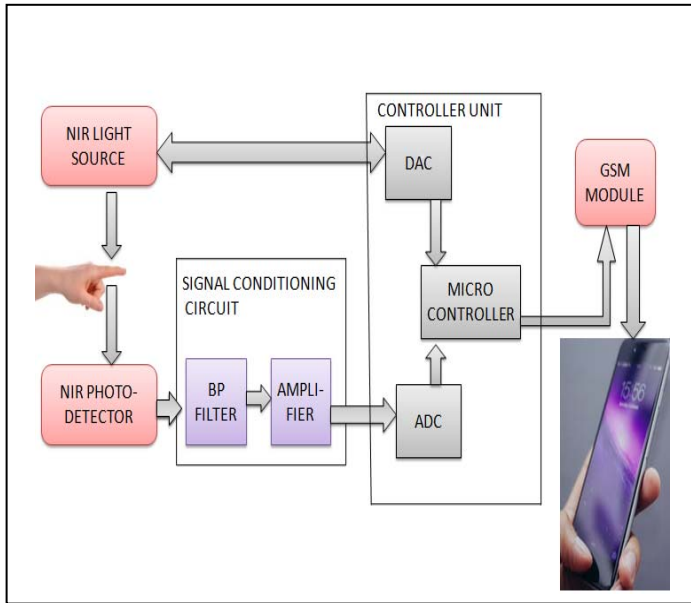


Fig. 3. Proposed block diagram.

III. MATHEMATICAL MODEL

The Beer-lambert law consists of two relations that, intensity of sample varied according with concentration and depth of path. An exponential decrease of transmitted light occurs with increase in concentration and distance covered increased. Transmittance depends on absorbance and optical distance to be covered.

$$A = \epsilon c l \quad (1)$$

A-Absorbance

c- Concentration

l- Length of sample

ϵ - Expectation coefficient

$$T = 10^{-A} \quad (2)$$

$$A = \log_{10}(I_0/I) = \epsilon c l \quad (3)$$

I_0 -Received intensity

I-Transmitted intensity

IV. RESULT AND ANALYSIS

In this paper, we present a of glucose measurement using non-invasive system using IoT. The work is under progress. We are currently working to validate the system using software and hardware. Following are the area that covers the results and analysis related works of this project.

- To test the closeness value of proposed system with conventional measured value.
- Performance of proposed system with existing system.
- Reading with different wavelength to analyse accuracy and resistance to noise.
- Absorption spectra analysis under different glucose concentration.
- Comparison of results between commercial invasive sensor and our proposed non-invasive sensor.

V. CONCLUSIONS

To design a optical based non-invasive glucometer with IoT for the diabolic patients for painless sugar level measurement.

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